

Course Syllabus - ChE 354 Transport Processes (Unit Operations 1)

Fall 2026 (Unique #s 17095-17100)

Course Description: This course covers analysis and design of fluid flow equipment, heat transfer equipment, and some interphase contacting devices. It serves to integrate the fundamental concepts learned in ChE 319, 210 and 317 with practical chemical engineering problems and commercial applications. Key objectives of this course are to enhance the student's problem-definition and problem-solving skills, and to improve the student's ability to perform calculations and obtain correct final answers for common commercial and industrial problems. Process safety concepts are also introduced.

Prerequisites: CHE 319.

Lectures: MWF 11:00 AM – 12:00 PM CPE 2.212 (17095)
MWF 12:00 PM – 01:00 PM CPE 2.210 (17100)

Recitations: W 03:00 PM – 05:00 PM GLT 5.104 (17095 + 17100)

Instructors: Dr. Thomas Badgwell / tbadgwell@che.utexas.edu /
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Teaching Assistants: Jason Tillman / jh83564@my.utexas.edu /
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Tutors: Prince Arwan / pfa287@my.utexas.edu /
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Course website: on Canvas <http://canvas.utexas.edu>

Req. Text: *Unit Operations of Chemical Engineering*, 7th Edition. McGraw-Hill, (2005), available through the Longhorn Textbook Access (LTA) program.

Req. Calculator: One able to solve multiple nonlinear equations, e.g. TI-Nspire or HP-Prime.

Schedule: Note that this schedule may change slightly as we work our way through the semester. The instructors will provide updates at the beginning of each lecture.

Week #	Week	Chapter(s)	Mo Lecture Topic	We Lecture Topic	We Recitation Topic	Fr Lecture Topic	Due Sat. 11:59 PM
1	24-Aug-2026	1/2	Intro/Syllabus	L1	L1	L2	HW1
2	31-Aug-2026	2/3/4	L3	L4	L2,L3, L4 (Q1)	L5	HW2
3	7-Sep-2026	4	LD-Holiday	L5	L4,L5 (Q2)	L6	HW3
4	14-Sep-2026	5	L7	L8	L6,L7,L8 (Q3)	L9	HW4
5	21-Sep-2026	8	L10	L10	L9,L10 (Q4)	L11	HW5
6	28-Sep-2026	8/7	L12	L13	L11,L12,L13 (Q5)	L13	HW6
7	5-Oct-2026	6	L14	L15	L13,L14,L15 (Q6)	L16	HW7
8	12-Oct-2026	1/2/3/4/5/6/7/8	Exam 1 Review		Exam 1	L17	
9	19-Oct-2026	6/8/10	L18	L19	L17,L18,L19 (Q7)	L20	HW8
10	26-Oct-2026	10/12	L21	L22	L20,L21,L22 (Q8)	L23	HW9
11	2-Nov-2026	11/15	L24	L25	L23,L24,L25 (Q9)	L26	HW10
12	9-Nov-2026	15/14	L27	L28	L26,L27,L28 (Q10)	L28	HW11
13	16-Nov-2026	14	Exam 2 Review		Exam 2	L29	
14	23-Nov-2026			Fall Break			
15	30-Nov-2026	10/11/14/15	L29	L30	L30 (Q11)	L30	HW12
16	7-Dec-2026		Final Exam Rev.			Final Ex	
17	14-Dec-2026						

Lecture Number	Lecture Topic	Lecture Number	Lecture Topic
L1	Units and Dimensions	L16	Adiabatic Pipe Flow
L2	Fluid Statics	L17	Isentropic Flow
L3	Decanters	L18	Normal Shock Behavior
L4	Macroscopic Mass Balances	L19	Compressors
L5	Macroscopic Momentum Balance	L20	Heat Transfer - Introduction
L6	Macroscopic Mechanical Energy Balance	L21	Time-Position Dependent Conduction
L7	Friction Losses - Conduits	L22	Convective Heat Transfer
L8	Other Friction Losses	L23	Overall Heat Transfer Coefficient
L9	Non-Newtonian Fluids	L24	Heat Exchanger Temperature Profiles
L10	Pumps	L25	Shell and Tube Heat Exchangers (SATHE)
L11	Flow Measurement	L26	SATHE - Design
L12	Flow Around Immersed Objects	L27	SATHE - Outside Heat Transfer Coefficient
L13	Fixed And Fluidized Beds	L28	SATHE - Basic Design Algorithm
L14	Compressible Flow	L29	Radiant Heat Transfer
L15	Isothermal Compressible Flow through Pipes	L30	HEN Design & Pinch Analysis

		Color codes:	Prof. Badgwell	Prof. Allen			
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Course Score (100 points total):

Exam Average (Two Midterms and a Final, 20% each)	60%
Homework Average (12 HWs)	25%
Quiz Average (best 10/11)	10%
Class Participation	5%

Course Grade (Course Score will be rounded to the nearest integer to determine grade):

90%	<= A-	< 94%	<=	A	<=	100%
80%	<= B-	< 84%	<=	B	< 87%	<= B + < 90%
70%	<= C-	< 74%	<=	C	< 77%	<= C + < 80%
60%	<= D-	< 64%	<=	D	< 67%	<= D + < 70%
				F	<	60%

Exams: Two Midterm Exams are scheduled, one on Wednesday, October 14th, and one on Wednesday, November 18th. Each exam will be two hours long, and will take place during recitation. Note that there will be no classroom lectures on exam days. The Final Exam will take place Friday, December 11 from 3:30-6:30 PM in GLT 5.104. If you have an unavoidable conflict and need to arrange a different time for an exam, you must notify the instructors at least one week in advance.

Homework: Homework will be assigned each week on Saturday, and due the following week on Saturday by midnight. Extensions may be granted in extraordinary circumstances, but only prior to the deadline. Unless an extension is granted, late homework turned in within one week of the deadline will still receive 80% credit. You are responsible for verifying that your homework has uploaded properly to Canvas. You are encouraged to take a first pass on the homework problems yourself, and then work with your fellow students to converge on problem solutions. In any case, what you turn in must be your own work.

Quizzes: Quizzes will be given at the end of each recitation period in order to test understanding of the fundamentals. Students will be able to drop one quiz from the final quiz average.

Class participation: During regularly scheduled class periods, various types of unannounced class activities will be conducted that will result in materials being turned in at the end of the lecture. These activities will be graded based only on whether the activity was completed and turned in. If class attendance is not possible for some reason, participation passes may be granted, but only in advance. Students will be able to drop one class activity grade from the final class participation average.

Knowledge, abilities and skills students should have upon entering this course:

1. Familiarity with engineering units and how to convert from one system of units to another. The ability to work exclusively in assigned unit systems (SI, CGS, US Customary) without converting to other systems is assumed.
2. Ability to write macroscopic mass, momentum, and energy balances under steady state and dynamic conditions (from ChE 210, ChE 317 and 319).
3. Familiarity with using equations of state including the ideal gas law (from ChE 317).
4. Familiarity with concepts of enthalpy, work, heat, and other thermodynamic quantities (from ChE 317).

5. Knowledge of the fundamentals of transport phenomena including familiarity with the equations used to describe momentum and heat transfer (from ChE 319).

Knowledge, abilities, and skills students should gain from this course:

1. Familiarity with the methods used to analyze and design fluid flow and fluid processing equipment, heat transfer equipment, and some interphase contacting devices as well as some of their operating characteristics.
2. Recognition of the challenges of designing processing equipment for compressible fluids and the ability to use the governing equations in flow and processing applications.
3. The ability to analyze and size piping networks, valves, pumps, compressors, and flow meters.
4. The ability to determine heat transfer coefficients and the ability to size and rate heat exchangers.
5. The ability to design a network of heat exchangers that minimizes the utility requirements of a process.
6. The ability to apply the principles of momentum and heat transport to the analysis of a wide range of processing equipment including evaporators, separation equipment, fluidized beds, and chemical reactors.
7. Introduction to safety including topics related to process equipment design and operation.

Matlab Software: Please download the latest MATLAB software and install it on your own computer. Matlab is available free-of-charge to UT students at:

<https://www.mathworks.com/academia/tah-portal/university-of-texas-at-austin-875352.html>

Communication: Email will be the primary means of communication with members of the instruction team. **Please preface your email subject line with “CHE 354 –” so that we may sort your email appropriately.** Please email us directly, rather than sending a message through Canvas (Replies to Canvas emails bounce consistently).

Classroom Engagement: Students are expected to attend classes, pay attention, and participate in class discussions. Phones should be stowed in backpacks. Studying material from other classes or doing homework during lectures is frowned upon. Remember – the first requirement of any job is to show up and be engaged.

Professionalism Statement: In the classroom you are expected to exhibit the same level of courtesy, sensitivity, and respect for others that you would find in a professional workplace environment. Courtesy and sensitivity are especially important in a group setting where you may interact with people of different cultures, religions, politics, sexual orientations, genders, gender variances, and nationalities. The goal is for everyone to feel comfortable attending this class, participating in discussions, and asking questions.

Feedback: Your feedback is important, and you are encouraged to let us know right away if you have strong feelings about anything that we do, positive or negative. Our goal is to teach the best possible course and we believe strongly that there is always room for improvement. You'll have two scheduled opportunities to provide feedback, via the Midterm Course Survey and the final Course Evaluation. Your responses to these will be voluntary, however we encourage you to provide constructive feedback that will allow us to improve the course. For both the midterm survey and the final Course Evaluation, if a class participation rate of 85% is achieved averaged over both Sections of the course, the entire class will receive credit for one classroom engagement activity.

Recitations: The Teaching Assistants will conduct recitations, where quizzes will be given, important concepts will be reviewed and problems like those on the HWs will be discussed and solved.

Regrades: All regrade requests for homework, quizzes, and exams must be submitted by email within one week of receiving the grade. Homework regrade requests should be directed to the grader, with the TAs CC'd. Quiz regrade requests should be directed to the TAs. Exam regrade requests should be directed to the instructors.

Electronics: At the start of class, please set your cell phone to silent and put it away. You may bring a laptop or iPad to lectures to take notes.

AI technology: Use of Artificial Intelligence technology, including Large Language Models such as ChatGPT, on assignments and exams is prohibited. We may, however, occasionally use these tools in lectures and homework assignments.

University of Texas Honor Code: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community.

Student Recording of Lectures: University regulation (HOP 2-9970) prohibits students from recording class instruction (audio or video) unless a student obtains the instructor's permission or Disability & Access has approved audio recording as an accommodation.

Cheating: Students who are caught cheating will be reported immediately to the office of Student Conduct and Academic Integrity without hesitation or prior warning. The referral will be accompanied by a recommended sanction that may be severe, such as a one-letter grade reduction in the final course grade. Once accused of academic misconduct in a course you cannot drop that course. Please be proactive about avoiding the appearance of cheating. **Please note that exams are scanned when they are turned in, and also after they have been graded.**

Disability Statement: The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let the instructors know if you experience any barriers to learning so we can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, please contact Services for Students with Disabilities (SSD). Please refer to SSD's website for contact and more information: <http://diversity.utexas.edu/disability/>. If you

are already registered with SSD, please deliver your Accommodation Letter to the instructors **within the first week** of the semester so we can discuss your approved accommodations and needs in this course.

Behavior Concerns Advice Line (BCAL): If you are worried about someone who is acting differently, you may use the Behavior Concerns Advice Line to discuss by phone your concerns about another individual's behavior. This service is provided through a partnership among the Office of the Dean of Students, the Counseling and Mental Health Center (CMHC), the Employee Assistance Program (EAP), and The University of Texas Police Department (UTPD). Call 512-232-5050 or visit <http://www.utexas.edu/safety/bcal>.

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have explicit, written permission. Unauthorized sharing of materials promotes cheating, and is a violation of the University's Student Honor Code and an act of academic dishonesty. The instructors are aware of sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to the office of Student Conduct and Academic Integrity.

Class Recordings: Class recordings are reserved only for students in this class for educational purposes and are protected under the Family Educational Rights and Privacy Act (FERPA). The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.